

## INTRODUCTION TO RADIOMICS

Marius E. Mayerhoefer<sup>1,2</sup>, Andrzej Materka<sup>3</sup>, Georg Langs<sup>2</sup>, Ida Häggström<sup>4</sup>, Piotr Szczypiński<sup>3</sup>, Peter Gibbs<sup>1</sup>, and Gary Cook<sup>5,6</sup>

<sup>1</sup> Department of Radiology, Memorial Sloan-Kettering Cancer Center, New York, USA

<sup>2</sup> Department of Biomedical Imaging and Image-guided Therapy, Medical University of Vienna,  
Austria

<sup>3</sup> Institute of Electronics, Lodz University of Technology, Lodz, Poland

<sup>4</sup> Department of Medical Physics, Memorial Sloan-Kettering Cancer Center, New York, USA

<sup>5</sup> Cancer Imaging Department, School of Biomedical Engineering and Imaging Sciences, King's College  
London, London, UK

<sup>6</sup> King's College London & Guy's and St Thomas' PET Centre, St Thomas' Hospital, London, UK

### Correspondence:

Dr. Marius Mayerhoefer

Department of Radiology

Memorial Sloan-Kettering Cancer Center

1275 York Ave, New York, NY 10065

Tel.: +43-1-40400-48180; Fax: +43-1-40400-48980

Email: [mayerhom@mskcc.org](mailto:mayerhom@mskcc.org)

ORCID ID: 0000-0001-8786-8686

## **ABSTRACT**

Radiomics is a rapidly evolving field of research concerned with the extraction and quantification of patterns – the so-called radiomic features – within medical images. Radiomic features capture tissue and lesion characteristics such as heterogeneity and shape, and may, alone or in combination with demographic, histological, genomic or proteomic data, be used for clinical problem-solving. The goal of this CE article is to provide an introduction to the field, covering the basic radiomics workflow: feature calculation and selection, dimensionality reduction, and data processing. Potential clinical applications in nuclear medicine that include PET radiomics-based prediction of treatment response and survival will be discussed. Current limitations of radiomics, such as sensitivity to acquisition parameter variations, and common pitfalls will also be covered.

**Key words:** radiomics; artificial intelligence; machine learning; positron emission tomography; single photon emission tomography

### **Learning objectives:**

- (1) To describe the standard radiomics workflow
- (2) To know the basic concepts of radiomic feature classes
- (3) To recognize common limitations and pitfalls in radiomics studies

## INTRODUCTION

Radiomics is a heavily discussed topic in nuclear medicine, and medical imaging in general. While the term is not strictly defined, radiomics generally aims to extract quantitative, and ideally, reproducible information from diagnostic images, including complex patterns that are difficult to recognize or quantify for the human eye. (1,2) There are several scenarios where radiomic features extracted from metabolic imaging techniques such as PET and SPECT may be useful.

First, radiomics may be used to capture tissue and lesion properties such as shape and heterogeneity, and, on serial imaging, their changes over time, i.e. during treatment or surveillance. In oncology, assessment of tissue heterogeneity is of particular interest: genomic analyses have demonstrated that the degree of tumor heterogeneity is a prognostic determinant of survival, and an obstacle to cancer control (3-6). Studies have suggested that radiomic features are strongly correlated with heterogeneity indices at the cellular level (7,8). While biopsies only capture heterogeneity within a small tumor portion, and usually, just a single anatomic site, radiomics captures heterogeneity across the entire tumor volume. Unsurprisingly, radiomic features are therefore also associated with tumor aggressiveness (9). Radiomic features have also been suggested to predict clinical endpoints such as survival and treatment response, and, to be directly linked to genomic, transcriptomic, or proteomic characteristics (1,2,9). While even individual radiomic features may correlate with genomic data or clinical outcomes, the impact of radiomics is increased when the wealth of information that it provides –typically hundreds of features, a fraction of which will contribute to a disease-specific “radiomic signature”– is processed using machine learning techniques (10,11).

Second, radiomic data are mineable, meaning that, in sufficiently large datasets, they may be used to discover previously unknown markers and patterns of disease evolution, progression, and treatment response. This so-called “population imaging” approach (12) may either utilize unstructured data from different modalities (e.g. PET, CT and MRI) that were acquired for a specific but possibly unrelated

diagnostic purpose in broadly-defined groups; or, for instance in the German National MRI Cohort Study, may utilize a single imaging test in a large cohort for a multicentric longitudinal observational study (13). Such radiomic data can be combined with clinical, laboratory, histological, genomic, or other data, using unsupervised machine learning.

Since a major obstacle for radiomics research is the lack of communication between physicians and computer scientists –in particular, lack of a “common language”– this article aims to provide an introduction to the technical radiomics terminology for physicians. We elected not to include mathematical equations, but to use words in combination with illustrations to highlight key concepts,. Like any other technique, radiomics has technical limitations –including susceptibility towards image acquisition and reconstruction parameters– and pitfalls, which will also be. Finally, we will discuss examples of current radiomics applications in nuclear medicine.

## **RADIOMIC FEATURES CLASSES**

Knowledge of the different types of radiomic features is not *mandatory* for reading research papers on radiomics, and not even for performing radiomics studies. However, familiarity with core principles may facilitate interpretation of results and pre-selection of features for specific applications. The below overview only includes the most frequently encountered radiomic feature classes. For a more complete listing of radiomic features and underlying equations, we recommend the recently published Image Biomarker Standardisation Initiative whitepaper (14).

Radiomic features can be roughly subdivided into (15):

- (1) Statistical, including histogram-based and texture-based;
- (2) Model-based;
- (3) Transform-based; and

#### (4) Shape-based.

While radiomic features can be extracted from 2-dimensional regions of interest (2D ROIs) or three-dimensional volumes of interest (3D VOIs), we chose “ROI” as an umbrella term for both to improve readability. Also, statistical features can be calculated for unmodified or “discretized” gray-level intensities.. Neither gray-level discretization (where the range of gray-level values is reduced to a predefined number to improve robustness and reproducibility) nor feature value aggregation (used to obtain a single value when the same feature is calculated in different variations, in its simplest form by arithmetic means), are covered, as this would exceed the scope of this article.

### Histogram Features

The simplest statistical descriptors are based on the “global” gray-level histogram, and include gray-level mean, maximum, minimum, variance, and percentiles (14,15). Because these features are based on single pixel/voxel analyses, they are called first-order “texture” features. For PET, the commonly used SUVmax, SUVmean, and SUVpeak fall into this category. More sophisticated features include Skewness and Kurtosis, which describe the shape of the intensity distribution of data: Skewness reflects the asymmetry of the data distribution curve to the left (negative skew, below the mean) or right (positive skew, above the mean); whereas Kurtosis reflects the “tailedness” of a data distribution relative to a Gaussian distribution due to outliers. Other features include histogram Entropy and Uniformity (also called “Energy”) –notably, these differ from their co-occurrence matrix counterparts of the same name.

### Texture Features

*Absolute Gradient.* A simple approach to “true” radiomic texture description is the analysis of the absolute gradient, which reflects the degree or abruptness of gray-level intensity fluctuations across an image. For two adjacent pixels/voxels, the gradient is highest if one is black and the other one white,

whereas, if both pixels are black (or both are white) the gradient at that localization is zero. Whether the gray-level value increases from black to white (positive gradient) or decreases from white to black (negative gradient) is irrelevant for the gradient magnitude. Similar to histogram features, gradient features include gradient mean, variance, Skewness, and Kurtosis (Fig. 1) (14,15).

*Gray-level Co-occurrence Matrix (GLCM).* First described by Haralick et al., the The GLCM is a second-order gray-level histogram. GLCM captures spatial relationships of *pairs of pixels/voxels* with predefined gray-level intensities, in different directions (horizontal, vertical, or diagonal for a 2D analysis, or 13 directions for a 3D analysis), and with a predefined distance between the pixels/voxels (16) (Fig. 2). GLCM features include Entropy (Fig. 2), a measure of gray-level inhomogeneity/randomness; Angular Second Moment (also called Uniformity or Energy), which reflects gray-level homogeneity or order; and Contrast, which emphasizes gray-level intensity differences between pixels/voxels belonging to a pixel/voxel pair (14-16).

*Gray-level Run-length Matrix (GLRLM).* The GLRLM, described by Galloway et al. provides information about the spatial distribution of runs of consecutive pixels with the same gray-level value, in one or more directions, in 2D or 3D (17). GLRLM features include Fraction, which assesses the percentage of pixels/voxels within the ROI that are part of the runs and therefore reflects “graininess”; Long and Short Run Emphasis (Inverse) Moments (Fig. 1), which are weighted towards the presence of numbers of long and short runs, respectively; and Gray-Level and Run-Length Non-Uniformity, which assess the distribution of runs over different gray-levels and run lengths, respectively (14,15,17).

*Gray-level Size Zone Matrix (GLSZM) and Distance Zone Matrix (GLDZM).* Described by Thibault et al., the GLSZM is based on a similar principle to the GLRLM, but here, counts of the number of groups (so-called “zones”) of interconnected neighboring pixels/voxels with the same gray-level value form the basis for the matrix (18) (Fig. 2).. A more homogeneous texture will result in a wider and flatter matrix. GLSZM is not computed for different directions, but may be computed for different pixel/voxel distances that define the neighborhood. GLSZM features may be calculated in 2D (eight neighboring pixels) or 3D

(26 neighboring voxels), and, following GLRLM definitions, include Fraction (percentage of pixels/voxels that are part of the zones), Large and Small Zone Emphasis, etc. (14,18)

As a variation of the GLSZM, the distance zone matrix (GLDZM) not only assess zones of interconnected neighboring pixels/voxels with the same gray-level value, but requires them to have the same distance from the ROI edge. GLDZM features are therefore “hybrids” between a texture features and morphological features. The latter is also reflected by some self-explanatory GLDZM feature names, such as Small Distance High Gray-level Emphasis (18).

*Neighborhood Gray-tone Difference Matrix (NGTDM).* Proposed by Amadasun and King, the NGTDM quantifies the sum of differences between the gray-level value of a pixel/voxel and the mean gray-level intensity of its neighboring pixels/voxels within a predefined distance (19). Key features include NGTDM coarseness, busyness, and complexity. Coarseness reflects the gray-level difference between the central pixel/voxel and its neighborhood, and thus captures the spatial rate of changes in gray-level intensities; i.e., an ROI consisting of larger areas with relatively uniform gray-level values (i.e., a lower rate of spatial intensity changes) will have a high coarseness value. Busyness, on the other hand, reflects rapid gray-level value changes between the central pixel/voxel and its neighbors (i.e., a high spatial frequency of intensity changes), so that an ROI composed of many small areas with markedly different gray-level intensities will have higher busyness values.

*Neighborhood Gray-level Dependence Matrix (NGLDM).* Described by Sun and Wee, the NGLDM is also based on the gray-level intensity relationship between a central pixel/voxel and its neighborhood (20). Here, a neighboring pixel/voxel within a predefined distance is regarded as being connected to the central pixel/voxel if it meets the dependence criterion in terms of a defined range of gray-level value differences. The ROI is then analyzed for the presence of central pixels/voxels with intensity  $i$  and  $j$  dependent neighboring pixels/voxels. Again, similar to the GLRLM, NGLDM features include Large

Dependence Emphasis and Small Dependence Emphasis that reflect heterogeneity and homogeneity, and Gray-Level Non-Uniformity and Dependence Uniformity that reflect the similarity in gray-level values and similarity of gray-level dependencies throughout an ROI, respectively (14,20).

### **Model-based Features**

Model-based analyses aim to interpret spatial gray-level intensity information to characterize objects or shapes. A parameterized model of texture generation is calculated and fitted to the ROI; its estimated parameters are used as radiomic features (15). The autoregressive model is an example of a model-based approach, and is based on the idea that the gray-level intensity of a pixel is a weighted sum of the gray-level intensities of four neighboring pixels: the pixel to its left (Theta 1); top left (Theta 2); top (Theta 3); and top right (Theta 4). In addition, Sigma, which carries information about the variance of the minimum prediction error, measures texture regularity (15).

Fractal analysis also yields features that can be utilized for radiomics, in particular Fractal Dimension, which reflects the rate of addition of structural detail with increasing magnification, scale or resolution, and therefore serves as a measure of complexity. Lacunarity, a feature measuring the lack of rotational or translational invariance, reflects inhomogeneity (21).

### **Transform-based Features**

Transform-based methods including Fourier, Gabor, and Haar wavelet transforms analyze gray-level intensity patterns in a different space.. The discrete Haar wavelet transform, for instance, analyzes the frequency content of an image at different scales (15). Wavelet “decomposition” of an image is possible by applying a pair of so-called quadrature mirror filters, a high-pass and a low-pass filter (22). While the high-pass filter highlights the changes in gray-level intensity, and thus emphasizes image details, the low-pass filter smooths the image in terms of gray-level intensity, removing image details. Following signal decomposition, a set of spatially oriented frequency “channels” is available, which is



used to describe local image variability. The energies within the frequency channels are then used as features. High-pass filtering in both directions (HH, Fig. 1) captures diagonal details; high-pass filtering followed by low-pass filtering (HL) captures vertical edges; low-pass filtering followed by high-pass filtering (LH) captures horizontal edges; and low-pass filtering in both directions (LL) captures the lowest frequencies, at different scales, respectively (15). Notably, wavelet transformation cannot only be used for generation of radiomic features, but may also be used for image segmentation, or as a pre-processing step to texture analysis.

### **Shape-based Features**

Shape-based features describe geometric properties of ROIs. Many shape-based features are conceptually much simpler than other radiomic features, e.g. 2D and 3D diameters, axes, and their ratios. Surface- and volume- based approaches founded on the use of meshes (i.e., small polygons such as triangles and tetrahedral) are more complex. Features include Compactness and Sphericity, which describe how the shape of an ROI differs from that of a circle (for 2D analyses) or a sphere (for 3D analyses), and Density, which relies on the construction of a minimum oriented bounding box (or rectangle for 2D analyses) enclosing the ROI (14).

### **ACQUISITION PARAMETERS AND FEATURE STANDARDIZATION**

Image-derived metrics such as SUVs and radiomic features are sensitive to image acquisition settings, reconstruction algorithm, and image processing. Recently, Zwanenburg performed a meta-analysis on 42 PET radiomics studies to evaluate feature robustness, reproducibility and standardization; 21 were eligible for quantitative evaluation (23). Different aspects of data heterogeneity –caused by variations of acquisition parameter (e.g., scan duration, numbers of iterations and subsets, reconstruction type and algorithm, and spatial resolution) and image processing (segmentation method and gray-level intensity

discretization)— were investigated. Spatial resolution had the strongest effect with a coefficient of variation (CV) of 3.63; followed by scan duration (CV, 2.93); segmentation method (CV, 2.92); reconstruction method (CV, 2.30); Gaussian filter width for post reconstruction smoothing (CV, 2.23), number of iterations (CV, 1.81) and number of subsets (CV, 1.08). Segmentation also had a considerable effect (CV, 2.92).

Van Velden et al. reported a repeatability comparable to that of simple SUV measures for most radiomics features (60% having good reproducibility), and that features were more sensitive to delineation/segmentation than to the reconstruction method (24). Other studies also found radiomics repeatability similar to SUV repeatability (25), and a high sensitivity to image discretization (26). Lasnon et al. found that ordered subset expectation maximization (OSEM) with point-spread function (PSF) modeling and post-filtering produced comparable SUV and texture information compared to plain OSEM, but unfiltered PSF images showed higher heterogeneity (potentially more discriminative in stratifying patients) (27). Papp et al. compared the sensitivities of the individual PET radiomic feature classes to variations in spatial resolution and reconstruction algorithms, using sphere-based phantoms (28). To minimize feature variations, they recommended small voxel sizes, narrow Gaussian post-filtering, and, like Lasnon et al., the use of PSF modeling. Bin size had a small effect on radiomic features according to Yan et al., but number of iterations, post-filter width, and voxel size affected many of them (3, 8, and 35 of 61 features showed >20% CV) (29). Galavis et al. concluded that 80% of investigated texture features displayed large variability (>30%) as a result of varying acquisition and reconstruction parameters (2D or 3D OSEM algorithm, number of iterations, post-filter width, and voxel size) (30). Pfaehler et al. indicated that PET images reconstructed with PSF increased repeatability compared to images reconstructed with a standard OSEM or TOF algorithm (31).

Since spatial resolution appears to have a particularly strong effect on radiomic features, resampling of multicentric imaging data is a common strategy to ameliorate the effects of differences in resolution between scanners. Whybra et al. investigated the robustness of PET/CT radiomic texture and

shape features to trilinear and spline interpolation for resampling to isotropic voxel sizes (32). They reported that about two-thirds of the 141 radiomic features tested were robust to both resampling techniques, and another 21% potentially correctable. However, absolute differences between the two interpolation techniques were, in part, considerable, and thus, a single interpolation technique should be used consistently.

Recently, deep learning for PET reconstruction was proposed, either for post-processing of conventionally reconstructed images (33), within an iterative reconstruction framework (34), or to directly map PET data into images (35). These methods have been able to restore or reconstruct PET images of higher quality compared to conventional OSEM, making images less noisy without sacrificing resolution. Since noise (controlled by e.g. counting statistics and number of iterations) heavily influences calculated features, deep learning may possibly lead to more robust radiomic features.

<sup>18</sup>F-FDG

## **FEATURE HARMONIZATION, SELECTION AND REDUCTION**

### **Feature Harmonization**

Harmonization is a mathematical post-processing technique to remove the so-called batch effect (i.e., center-dependent effects of acquisition parameter variations) on radiomic features following image acquisition, reconstruction, and analysis. Harmonization is thus not applied to images, but directly to numerical radiomic feature values. The currently most popular technique is ComBat harmonization, which was originally described for use with genomic data and has meanwhile been validated for removing the center effect from radiomic features while preserving pathophysiological information (36). Several studies have applied this technique to PET radiomics (37,38).

## Feature Selection and Dimensionality Reduction

Once radiomic image analysis has been completed, the relevant features that will be used in the statistical model to solve the clinical problem (e.g., to distinguish between benign and malignant lesions) must be identified.

While theoretically, the hundreds of radiomic feature “candidates” that are typically extracted (Fig. 3) could be used as input to the prediction model, the number of required model parameters would then grow exponentially. Therefore, a large number of feature candidates must be removed or transformed—this is called *dimensionality reduction*. Radiomic features frequently show high correlations indicating data redundancy, meaning that some features can be discarded, and others grouped together and replaced with a representative feature, e.g., using principal component (PCA) or linear discriminant analysis (LDA). Among such representative features, “informative” features showing highest natural biologic range (i.e., inter-patient variability) are preferable (39). Fourteen approaches to radiomic feature selection were compared by Parmar et al., including mutual information-based methods, and twelve machine learning classifiers (40); a similar approach was also made by Leger et al, who focused on time-to-event survival data (41). Extensive radiomic feature selection systems for prediction of tumor treatment outcome from PET images were developed by Lian et al. (42).

Dimensionality reduction techniques that reduce redundancy without exploiting knowledge about target variables (e.g., benign or malignant), such as PCA, are popular but typically *mix* variables and render subsequent tracing of predictors in the initial radiomic feature set difficult. However, once prediction targets are taken into account during feature selection, care has to be taken to avoid so-called *overfitting* leading to overly optimistic estimates of

predictive accuracies (see below). This is because once the number of features is high enough, correlations can be detected even in random data.

## MODEL CONSTRUCTION AND CLASSIFICATION

Once the radiomic features have been selected, they are used to predict target variables

- (1) in the present, such as absence/presence of disease, or tumor type, or
- (2) in the future, such as treatment response , or time-to-recurrence

The target can either be scalar (e.g. survival in months), predicted by a *regression* model; or categorical (e.g., response status, receptor positivity), predicted by a *classification* model (43) (Fig. 3). Machine learning, a sub-field of artificial intelligence, has undergone rapid development during the last two decades. Machine learning models learn the relationship between high dimensional inputs –radiomic features– and target variables based on training examples. A key capacity is the exploitation of groups of predictors/features taken together, so-called *multivariate patterns* as opposed to *univariate* or *mass-univariate* regression. *Support vector machines* were one of the first highly successful models, suffering only from the need of careful feature selection (44). “Bagging” and “boosting” methods, such as *Random Forests* introduced the training of robust classifiers/regressors by integrating feature selection and effective sampling in the training process (45). This ability is at the core of many current radiomics approaches that select features based on their predictive contribution. They fostered extremely successful applications in various fields ranging from automotive (pedestrian recognition) to genetics (identifying associations), by shifting the focus from experts-based feature selection to directly letting the classifier evaluate features candidates and identify those with predictive value (46). More recently, neural networks have resurged as powerful classification and *feature construction* models. Given sufficient training data, *convolutional neural networks (CNN)* outperform feature selection schemes, because instead of selecting features from pre-defined and finite sets of feature candidates, they construct optimal features from the

image data itself (47). One limitation is the need for excessive training data that has recently successfully been tackled by strategies such as transfer learning, exploiting the similarity of visual features across problem domains.

Goodness-of-fit on the test-set is a valid measure of how well the approximation function predicts the output of the target function. For categorical variables, false positives, false negatives or derived measures reflect the relevant information. Since many of the approaches have parameters, receiver-operator-characteristics curves (ROC) are used to represent prediction accuracy over a parameter range. Cross-validation schemes, where training and test sets are rotated through the available data can alleviate limited dataset sizes but have to be used with caution. Once the algorithm design and its parameters are informed substantially by intermediate cross-validation, the ability to provide independent test results is lost. Here, a hold-out set of data that is only touched after algorithm finalization during a final validation is best practice. For further information about machine learning, please refer to the recent article by Uribe et al (11).

## **PITFALLS AND QUALITY CONTROL**

There are several *caveats* when conducting radiomics research, some of which were briefly mentioned above. These include the need for high-quality, artifact-free images, preferably obtained with homogeneous image acquisition protocols and reconstruction techniques, and/or the use of appropriate correction before or after radiomic analysis in situations where this cannot be achieved (e.g., in multicentric retrospective studies); adequate size and completeness of datasets; and separate training and a validation datasets. Apart from these prerequisites for meaningful radiomics research, two pitfalls that frequently affect radiomics studies: *class imbalances* and *overfitting*.

### **Class Imbalances**

Outside of randomized clinical trials, class imbalances are common. Especially in retrospective studies using routine clinical data, it is seldom the case that the condition of interest has the same prevalence as the lack of this condition within a cohort. For instance, in patients with diffuse large B-cell lymphoma, 18F-FDG, bone marrow involvement is found in approximately 16% of patients (68). When evaluating 18F-FDG-PET radiomics for detection of bone marrow involvement, this imbalance in the percentage of patients with (16%), and those without marrow involvement (84%) must be taken into account when judging the model performance. A classifier that assigns all cases in the sample to the “no marrow involvement” group would have a seemingly decent accuracy of 84% but would be clinically useless as it would be unable to actually distinguish between involved and uninvolved bone marrow on PET (see also Supplemental Table 1). Therefore, not only overall accuracy, but also class-wise accuracy, or sensitivity/specificity should be reported.

### **Overfitting and Underfitting**

If a model is not sufficiently well-balanced in terms of function approximation, overfitting, or, to a lesser degree, underfitting may be encountered. Overfitting occurs when a model with a large number of input parameters or too many degrees of freedom can “memorize” data, so that not just the relevant, disease-specific features, but also features reflecting image noise and random fluctuations are included in the model (Supplemental Fig. 1). Such a model gives correct classification results at the data points provided to it during training, but its response is wrong for points outside the training dataset—the model unable to *generalize* information. To avoid overfitting, *regularization* needs to be applied to make the model function smooth, or the number of input features needs to be reduced, which decreases the number of required model parameters. *Validation* using a separate dataset helps in detecting overfitting: if during training the error on the training set decreases, but starts increasing on the validation dataset, the training needs to be stopped. Underfitting, on the other hand, occurs when a model is incapable of classifying data correctly in both training and validation datasets, for instance because it is overly simplistic. Here,

additional input data, or switch to a different model may be necessary.

### **Radiomics Score**

Lambin et al. developed a modality-independent radiomics quality score based upon 16 criteria that carry different weights; a maximum of 36 points that can be achieved (48). While the use of standardized image acquisition protocols according to published recommendations is among these criteria, it has, with one point, just a minor impact; software-based correction or harmonization techniques for multicentric data are not explicitly mentioned, possibly due to the publication date; instead, the use of a phantoms to assess variations in radiomic feature values is endorsed. With seven points, particular weight is put on a prospective design and trial registration, and, with up to five points, use of a validation dataset. With three points, the use of feature reduction techniques to reduce the risk of overfitting is also a relevant criterion. Notably, with two points each, assessment of the added value of the radiomics approach in comparison to the current gold standard (e.g., radiomics vs. image-based TNM stage) as well as clinical relevance and utility are important factors. The score also recommends combination of radiomic with clinical, molecular and genomic data (48).

## **CLINICAL APPLICATIONS**

Compared to the multitude of studies utilizing “traditional” PET metrics for outcome prognostication or tissue characterization, applications of texture, shape or histogram features are still rare in the literature.. In this section we will review selected articles to highlight promising clinical applications, and also discuss their limitations.

### **Radiogenomics: Linking Imaging Data to Biology**



Radiogenomics in non-small cell lung cancer (NSCLC) has attracted particular interest. Nair et al. investigated the association of quantitative 18F-FDG-PET/CT-based metabolic tumor volume (MTV) and histogram features with genomic data in NSCLC patients (49). Fourteen radiomic features and three principal components were correlated with gene expression for single genes and co-expressed gene clusters in a training dataset of 25 patients that underwent PET/CT prior to tumor resection. Four genes (LY6E, RNF149, MCM6, and FAP) were correlated with radiomic features and survival. Unusual for a radiomics study, the two test/validation cohorts confirming these associations were much larger (63 and 84 patients) than the training cohort. Yip et al. investigated associations between 18F-FDG-PET/CT radiomic features of 348 NSCLC patients (histogram, GLCM, GLRLM, GRSZM, NGTDM, and shape) 18F-FDG and epidermal growth factor receptor (EGFR) or Kirsten rat sarcoma viral (KRAS) mutations (50). and histologic subtype heterogeneity.. While eight texture features (and also SUV and MTV) were significantly associated with EGFR, and one GLCM feature even predictive of EGFR+ status, no feature was associated with KRAS mutation. The study was limited by the lack of a validation cohort, and use of PET/CT data from eight different scanners; while voxel intensities were resampled, voxel size was not. . 18F-FDG

18F-FET-PET radiomics based were used by Pyka et al. to differentiate between tumor grades in 113 high-grade glioma patients (51). A single PET/CT device was used, eliminating the need for resampling or harmonization. Four NGTDM features enabled differentiation between tumor grades III and IV; further improvement was achieved through combination with MTV. Notably, no validation set was used. 18F-FLT-PET radiomics was retrospectively evaluated by Lohmann et al. for prediction of isocitrate dehydrogenase mutation (IDH mut), a diagnostic marker not routinely obtained preoperatively (52). 56/84 patients were examined on a stand-alone PET scanner, and the remaining 28/84 patients on a PET/MRI device –i.e., practically all acquisition parameters (including resolution and reconstruction algorithm) differed between the two subgroups. Thirty-three features (histogram, GLCM, GLRLM, GLSZM, and shape) were extracted; 26/84 patients showed IDH mutations . To avoid overfitting, the

number of relevant features was therefore reduced to two, and 5- and 10-fold cross-validation was applied. Accuracies of up to 71%, but quite low sensitivities, were achieved, probably due to a combination of class imbalance and acquisition parameter heterogeneity.

<sup>18</sup>F-FDG In an experimental setup, Rajkumar et al. tested whether GLCM features extracted from [<sup>125</sup>I]-A5B7 anti-carcinoembryonic antigen antibody nano-SPECT could be used to differentiate between metastatic colorectal cancer phenotypes (53). In 14 mice with hepatic colorectal cancer metastases, the authors found that undifferentiated metastases were clearly more heterogeneous than well-differentiated lesions, as reflected by three SPECT texture features, which also captured antivascular therapy effects.

### **Clinical Outcome Prediction**

Early treatment response assessment and response and survival prediction is of interest to clinicians, as it may aid treatment selection and patient stratification, and justify a therapy switch. In 358 stage I-III NSCLC patients, Arshad et al. used pre-therapeutic <sup>18</sup>F-FDG data from seven institutions to evaluate <sup>18</sup>F-FDG-PET radiomics for overall survival (OS) prediction after radiotherapy or chemoradiotherapy (54). Histogram, shape, and texture features (GLCM, GLRLM, and NGTDM, extracted from original and wavelet-transformed images) were calculated in addition to traditional PET metrics, and dimensionality reduction by least absolute shrinkage and selection operator (LASSO) regression in combination with weighted linear feature combination was performed. No correction for acquisition parameter variations was applied. 133 datasets were used for training, 204 patients for internal validation, and 21 patients for external testing. The combined radiomic feature vector correctly predicted 14-month survival difference in the validation cohort, and lack of survival difference in the testing cohort.

Peng et al. evaluated a pre-treatment <sup>18</sup>F-FDG-PET/CT-based radiomics signature and nomogram to predict disease-free survival of patients with stage III-IVa nasopharyngeal carcinoma, using images obtained with a single scanner type and fixed acquisition protocol, to predict disease-free survival

(DFS) (55). <sup>18</sup>F-FDG. The training dataset consisted of 470 patients, and the validation set of 237 patients; Radiomic features were chosen manually (including histogram, shape, GLCM and GLRLM features), and then also automatically by deep-learning CNNs. LASSO Cox regression analyses were used for feature dimensionality reduction. The radiomics nomogram proved superior to nomograms based on clinical data and plasma EBV DNA (an established prognostic biomarker in nasopharyngeal carcinoma). The radiomics nomogram enabled patient stratification into two risk groups that differed in terms of 5-year DFS; only the radiomics high-risk group showed a benefit from induction chemotherapy in addition to standard chemoradiotherapy. Interestingly, no combination between radiomic features and other data (such as DNA) was evaluated. Such a strategy was, however, evaluated in a similar study by Lv et al. in 128 patients with nasopharyngeal carcinoma; here, the combination of radiomic and clinical data slightly improved prediction of progression-free survival (PFS) (56)).

<sup>18</sup>F-FDG Finally, in a study of 214 gastric cancer patients, Jiang et al. investigated the utility of <sup>18</sup>F-FDG-PET radiomic features (histogram, shape, GLCM, GLRLM, GLSZM, and NGTDM) for DFS and OS prediction (57). Here, the training cohort (132 patients) was examined with a single scanner, and the validation cohort (82 patients) with a different scanner from a different vendor, providing true external validation. Although voxel size and other acquisition parameters differed between the two cohorts, the radiomics score built on features selected through LASSO regression was a better predictor of OS and DFS than TNM stage or the tumor marker CA-19-9.. Again, no CT radiomic features or clinical/laboratory data were included in the radiomics prediction model.

## CONCLUSIONS

Radiomics is a sophisticated image analysis technique with the potential to establish itself in the precision medicine. Radiomic features are not only correlated with genomic data, but may provide complementary information about tumor heterogeneity across the entire tumor volume to improve

survival prediction and may therefore prove useful for patient stratification. For nuclear medicine with its long tradition of providing quantitative biological data, radiomics could represent the next logical step in its evolution, not just as a clinical decision-making tool, but also a research tool to discover novel molecular disease pathways. However, development of, and strict adherence to, standardized image acquisition and reconstruction protocols is vital

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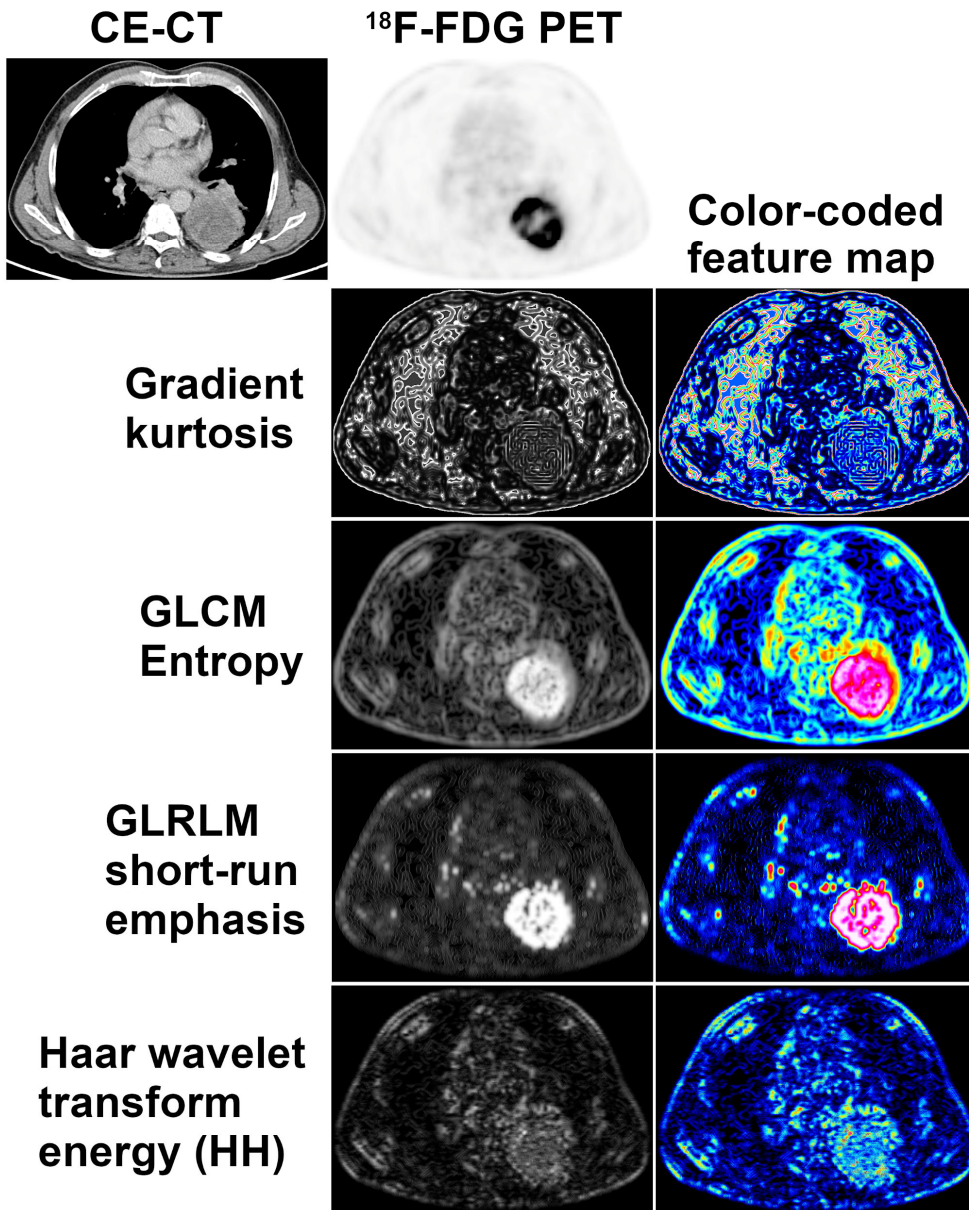
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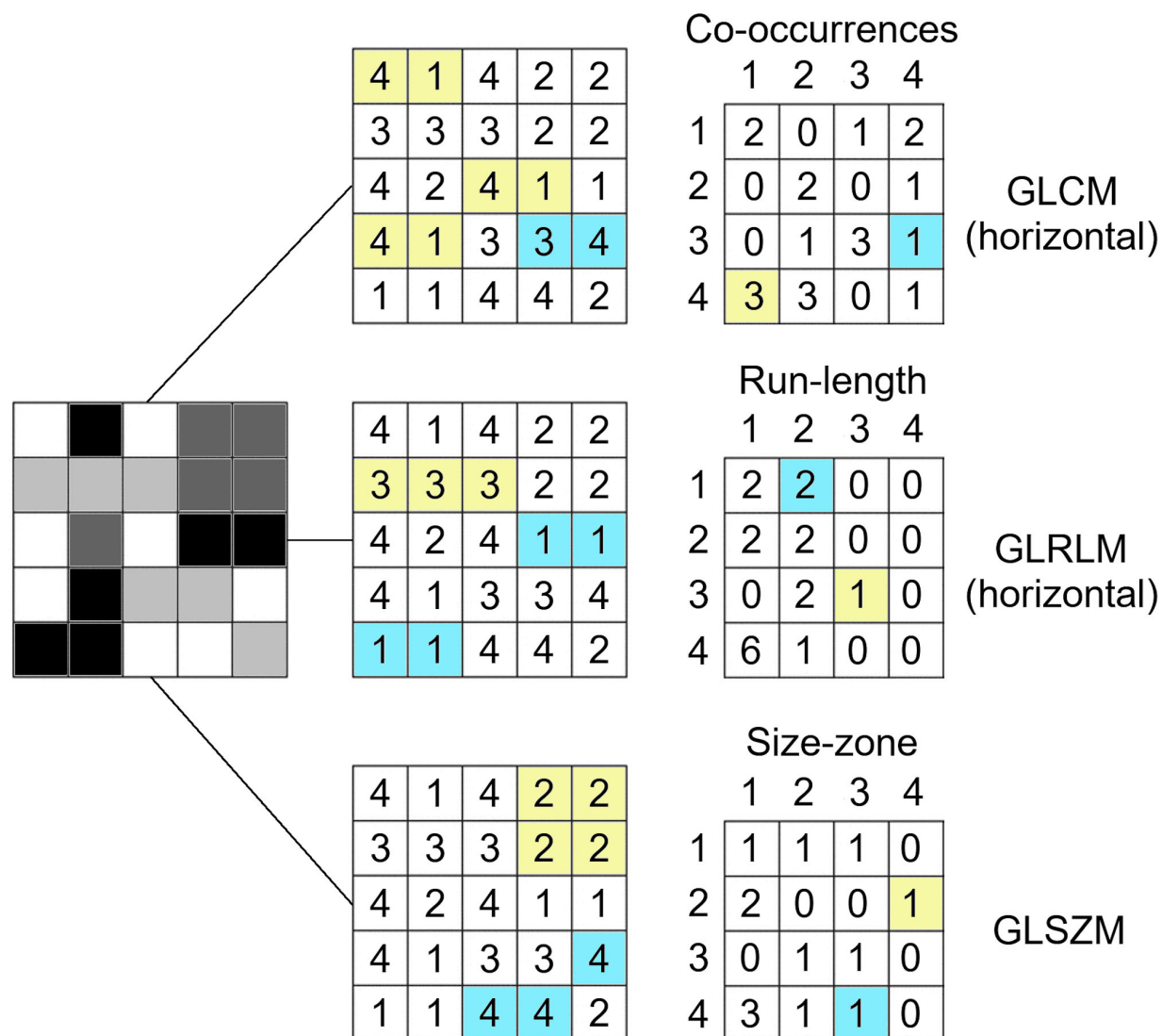


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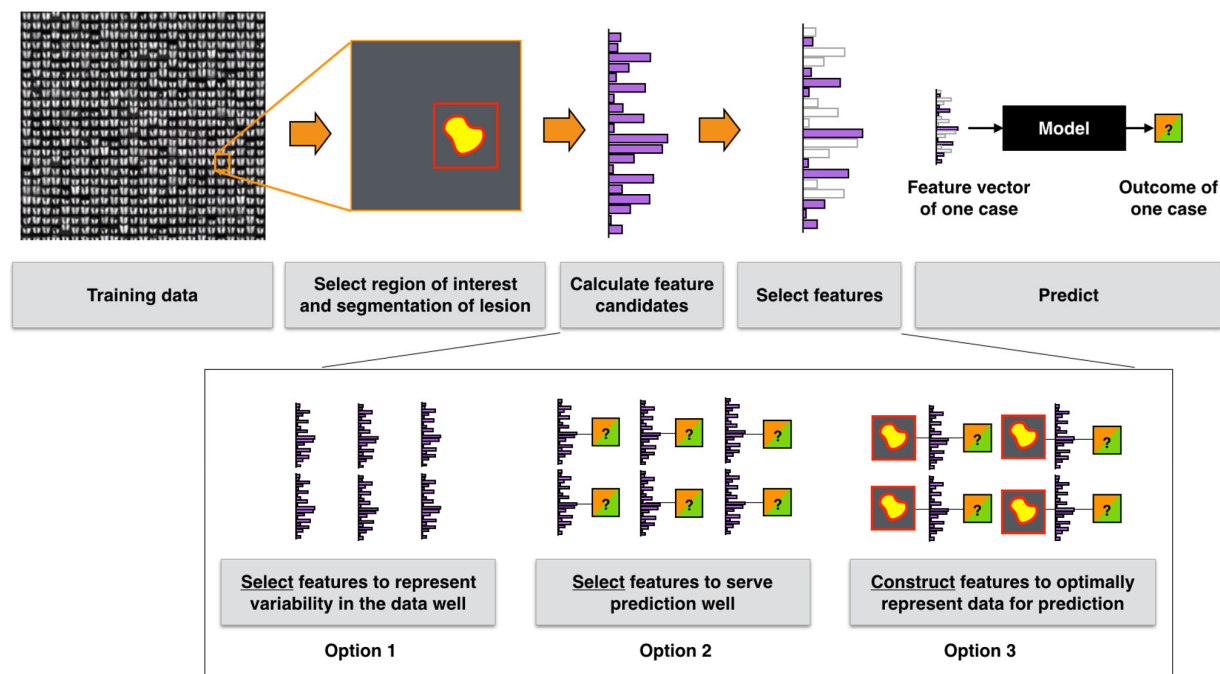
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**Fig. 1. Visual representation of radiomic features.**  $^{18}\text{F}$ -FDG-PET and contrast-enhanced CT images of a partly necrotic lung cancer of the left lower lobe. Radiomic feature maps generated by moving a small rectangular window over the PET image, and calculating a feature value for each position, reflect different aspects of glucose metabolism heterogeneity across the tumor. Each feature map depicts a single radiomic feature, with high values corresponding to high signal intensities on gray-level features map. Color-coded features maps may be used for better visualization and as a color overlay for CT.

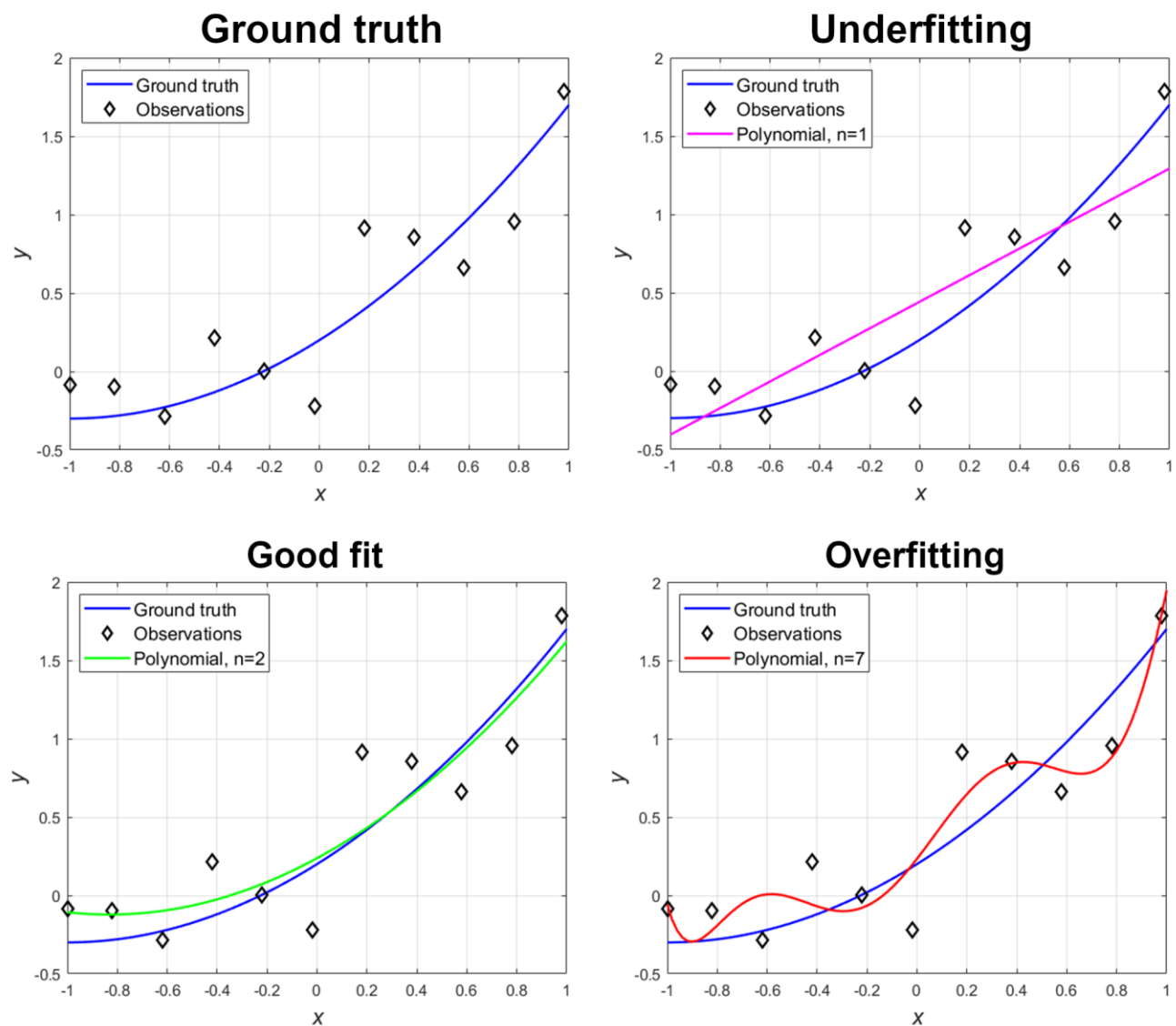


**Fig. 2. Calculation of radiomic texture features.** While the GLCM relies on pixel pairs (here: interpixel distance = 0), the GLRLM relies on runs, and the GLSZM on areas of neighboring pixels with the same gray-level value, respectively.



**Fig. 3. Radiomics workflow.** First, a region of interest is defined and/or lesions are segmented. For ROIs/lesions, a frequently large number of feature candidates are extracted. Features that either represent variability in the data most efficiently, or serve a particular prediction model best, are selected. Instead of selecting from a pre-defined set of features, deep learning approaches link feature construction and modeling directly to achieve further improvements of prediction accuracy and reliability.<sup>18</sup>F-FDG

**Supplemental Fig. 1. Underfitting and overfitting.** In search of a model that provides the best approximation of the target function, a simplistic model using a linear function leads to underfitting: neither the ground truth nor the noisy observations are correctly predicted. On the other hand, a very complex model using a 7th order polynomial function leads to overfitting: it “memorizes” the deviations from the ground truth (i.e., image noise), may therefore perform well in the training dataset, but will fail at external validation. The well-balanced model, in this case a quadratic function, which is neither too simplistic nor too complex, provides a good fit.



**Supplemental Table 1. Effects of class imbalance**

| <b>Prediction by classification model</b> |                     |                |                |                 |
|-------------------------------------------|---------------------|----------------|----------------|-----------------|
|                                           | <b>Ground truth</b> | <b>Class 1</b> | <b>Class 2</b> | <b>Accuracy</b> |
| <b>Training (n=80):</b>                   | <b>Class 1</b>      | 55             | 4              | 93.2%           |
|                                           | <b>Class 2</b>      | 13             | 8              | 38.1%           |
|                                           | <b>Overall</b>      | 85%            | 15%            | 78.8%           |
|                                           |                     |                |                |                 |
| <b>Validation (n=40):</b>                 | <b>Class 1</b>      | 22             | 2              | 91.7%           |
|                                           | <b>Class 2</b>      | 11             | 5              | 31.3%           |
|                                           | <b>Overall</b>      | 82.5%          | 17.5%          | 67.5%           |

While the overall accuracy in the training dataset is seemingly satisfactory with 78.8%, this is due to a class imbalance: since only 26% (21/80) of all cases belong to Class 2, the low accuracy of the prediction model for this class (38.1%) is masked by the high accuracy in Class 1 (93.2%) to which 74% of all cases belong. Since the validation dataset contains a higher percentage (40%) of Class 2 cases (16/40), overall accuracy is lower (67.5%), while classification accuracies within the two classes are comparable between training and validation datasets. This confirms that the performance of the model, which assigns >80% of all cases to Class 1, is unsatisfactory.



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Marius E Mayerhoefer, Andrzej Materka, Georg Langs, Ida Häggström, Piotr Szczypinski, Peter Gibbs and Gary Cook

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